WORKSHEET

Integration of power functions

1 Evaluate the following definite integrals.

a $\int_{2}^{5} x^2\,dx$

b $\int_{7}^{9} x^3\,dx$

c $\int_{0}^{3} -x^2\,dx$

d $\int_{1}^{2} -x^5\,dx$

e $\int_{1}^{3} x^4\,dx$

f $\int_{1}^{2} -x^4\,dx$

2 Find the values of the following definite integrals.

a $\int_{-1}^{1} x^2\,dx$

b $\int_{-2}^{4} x^2\,dx$

c $\int_{-3}^{0} x^2\,dx$

d $\int_{-2}^{-1} x^2\,dx$

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e $\int_{-2}^{4} -x^2\,dx$

f $\int_{-2}^{0} -x^2\,dx$

3 Evaluate the following definite integrals.

a $\int_{2}^{0} x^2\,dx$

b $\int_{-1}^{0} x^3\,dx$

c $\int_{-1}^{1} x^4\,dx$

d $\int_{-2}^{1} -x^5\,dx$

e $\int_{2}^{-1} -x^2\,dx$

f $\int_{5}^{3} x^4\,dx$

g $\int_{-1}^{-3} -x^2\,dx$

h $\int_{10}^{12} -x^3\,dx$

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Answers

1 **a** 39

b 1040

c -9

d -10.5

e 48.4

f -6.2

2 **a** $\frac{2}{3}$

b 24

c 9

d $\frac{7}{3}$

e -24

f $-\frac{8}{3}$

3 **a** $-\frac{8}{3}$

b -0.25

c 0.4

d 10.5

e 3

f -576.4

g $\frac{26}{3}$

h -2684

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